Mohammad Rico Ferdian Khoirunizar

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SUMMARY

A motivated and dedicated Junior Electrical Engineer with over 3 years of experience, spanning industrial maintenance (motors, panels, PLCs), power system analysis (ETAP, IEC 60909), and lithium-ion battery research, including the use of data acquisition (DAQ) systems and deep learning-based forecasting. Combines practical field expertise with data-driven analytical skills.

EXPERIENCE

Electrical Engineer Intern

PT Petrokimia Gresik **December 2025 - January 2026, Gresik, Jawa Timur**

- Successfully executed the replacement of a critical **6kV** motor in a high-risk area safely, optimally, and under operational pressure through solid team coordination.
- Contributed directly to the smooth and successful execution of the Factory Acceptance Test (FAT) for a **6kV 400kW** motor at ABB Surabaya.
- Analyzed short-circuit faults within the **6kV** electrical distribution network at transformer **TR-2281-B**.
- Developed and conducted power system simulations using ETAP to evaluate short-circuit fault conditions in accordance with **IEC 60909** standards.
- Analyzed short-circuit currents and evaluated fault conditions within an industrial distribution network to assess electrical system behavior, while supporting the evaluation of protection system coordination, reliability, and safety.

Electrical Technician

PT Wiharta Karya Agung **February 2022 - March 2023, Gresik, Jawa Timur**

- Executed single-phase and three-phase induction motor rewinding, handling motor ratings of **0.18kW, 0.25kW, 0.75kW, 1.5kW, 2.2kW, and 4kW**.
- Performing inspection, disassembly, and reassembly of motors as part of repair and maintenance activities.
- Conducting initial assessments of electric motors to identify damage and potential electrical or mechanical issues.
- Applying electrical safety procedures during motor rewinding and equipment maintenance.
- Performing regular inspections, routine maintenance, and troubleshooting on industrial motors, MV/LV panels, and PLC systems to minimize downtime and maintain optimal performance of electrical systems.

Air Conditioning Technician

ARTECH Service AC **June 2021 - January 2022, Gresik, Jawa Timur**

- Optimizing the performance of air conditioning systems through thorough cleaning, routine servicing, and preventive maintenance.
- Diagnosing technical issues by checking refrigerant pressure, electrical systems, and cooling components to identify malfunctions and carry out effective repairs.

PROJECT

Lithium-Ion Battery Research: Data Acquisition, Analysis, and Deep Learning Forecasting

Muhammadiyah University, Gresik - Doctoral Research Project • September 2025 - Present

- Contributed to a one-year doctoral research project focused on lithium-ion battery monitoring, data analysis, and performance forecasting.
- Assisted in developing and implementing data acquisition (DAQ) systems to collect and record battery operating data.
- Performed data preprocessing, analysis, and interpretation to identify battery performance and degradation patterns.
- Applied deep learning-based forecasting techniques to analyze battery behavior and predict future performance.
- Supported research activities involving battery testing, experimental data management, and evaluation of model performance.

EDUCATION

Electrical Engineering

Muhammadiyah University, Gresik • Jawa Timur, Indonesia • 2026 • GPA : 3.59

SKILLS

Knowledge: Electrical Power System, Motor, Renewable Energy, PLC & HMI (Siemens, Haiwell).

Technical: Measuring, Inspecting, Commissioning, Troubleshooting, Programming, Designing.

Software: ETAP, AutoCAD, MATLAB, TIA Portal, Haiwell Cloud SCADA, Python, Ms. Office.